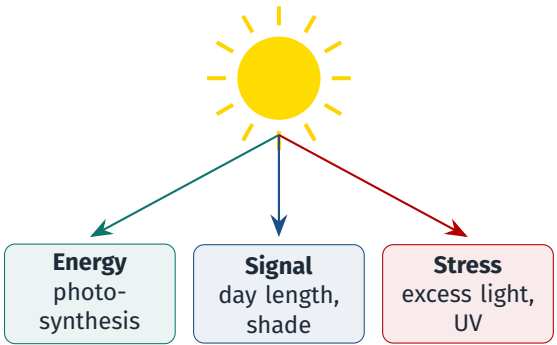
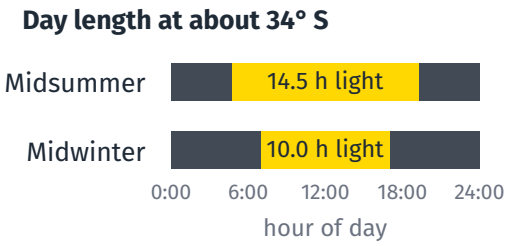
Light: a resource, a signal and a stress

- Light is the energy source for photosynthesis
- It is also a signal for timing development
- In excess, it damages the photosynthetic apparatus
- Atmosphere, canopies and water change light before it reaches a leaf or alga



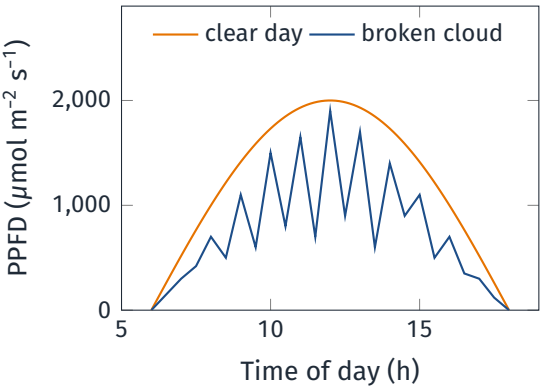
Why light matters to plants

- **Energy**: photosynthesis converts light energy to chemical energy
- **Signal**:
  - photoperiodism: response to day length (e.g. flowering)
  - circadian rhythms: internal clocks kept in time by light
  - photomorphogenesis: light controls growth and form
- **Stress**: too much light or UV damages photosystems
- With temperature, light largely sets where plants grow



Light varies in space and time

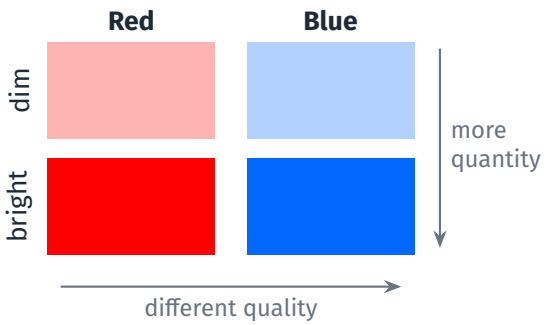
- **Predictable**: day length and Sun height change with latitude and season
- **Unpredictable**: cloud, aerosols, moving leaves, waves
- Annual totals vary little between years
- Minutes to days: light varies a great deal
- **PPFD** (photosynthetic photon flux density): photons of 400–700 nm per m^2 per s, in μmol m^{-2} s^{-1}



Schematic; not to scale.

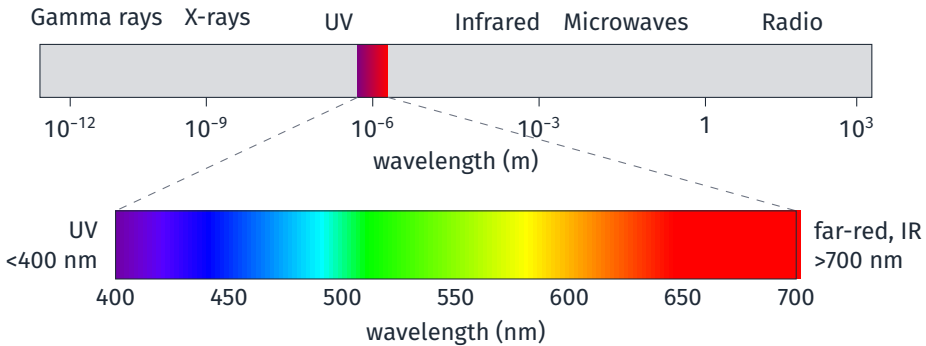
Light quantity and light quality

- **Quantity**: how much light, regardless of colour
- **Quality**: spectral composition, i.e. the relative amounts of each wavelength
- Quality includes UV and far-red, which we cannot see
- Quantity and quality vary independently



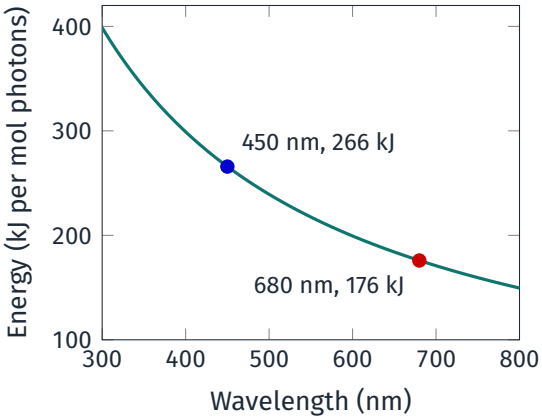
The electromagnetic spectrum

- Visible light is a narrow band, about 390–750 nm for the human eye
- Plants use a similar band for photosynthesis (PAR, 400–700 nm)
- UV lies at shorter wavelengths, infrared at longer wavelengths



Wavelength, frequency and photon energy

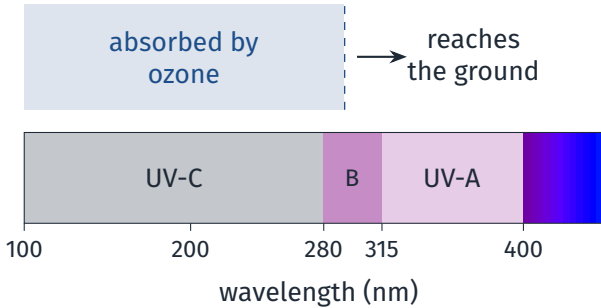
- $c = \lambda\nu$: short wavelength means high frequency
- Photon energy: $E = h\nu = hc/\lambda$
- Each blue photon carries more energy than each red photon
- This is per photon; total energy also depends on photon numbers
- 450 nm: 266 kJ mol^{-1}; 680 nm: 176 kJ mol^{-1}



Curve: $E = N_A hc/\lambda$.

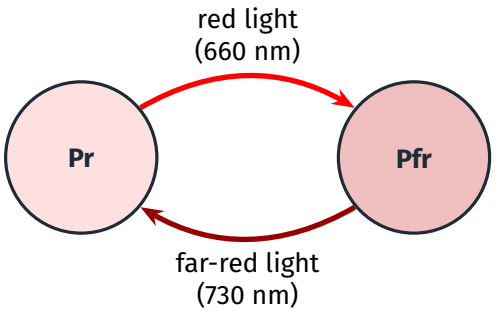
Ultraviolet radiation

- UV-C 100–280 nm, UV-B 280–315 nm, UV-A 315–400 nm
- Ozone removes UV-C and most UV-B; ground UV starts at about 290–295 nm
- Not ionising, but UV-B damages DNA and photosystem II (D1 protein)
- Plants sense UV (UVR8; cryptochromes, phototropins) and respond with flavonoid sunscreens



Infrared and far-red light

- About half of solar energy is infrared; beyond about 750 nm it acts mainly as heat
- Far-red (700–750 nm) is biologically active
- Phytochrome switches between a red-absorbing form (Pr, peak near 660 nm) and a far-red-absorbing form (Pfr, peak near 730 nm)
- Far-red adds to photosynthesis via photosystem I (Emerson enhancement)

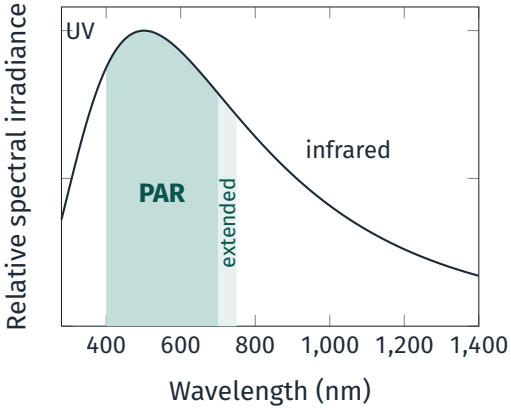


inactive form: favoured in canopy shade

active form: de-etiolation, flowering responses

Photosynthetically active radiation (PAR)

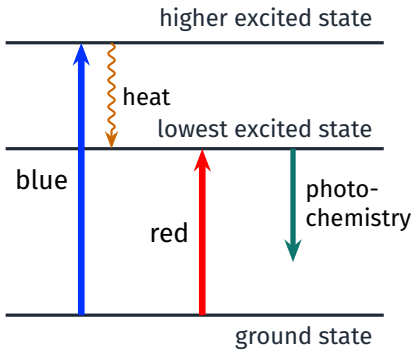
- **PAR = 400–700 nm** (McCree's definition)
- Quantum sensors measure this band
- The limits are a convention: some photons at 700–750 nm are also used ("extended PAR")
- PAR is roughly 45–50% of the energy in sunlight at the ground



Curve: 5,778 K black body, a schematic of sunlight without atmospheric absorption.

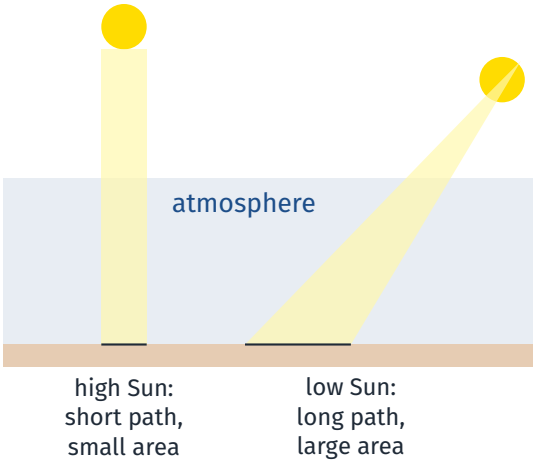
Photochemical equivalence

- One absorbed photon excites one electron, whatever its wavelength (Stark–Einstein law)
- Blue light: higher excited state, then a drop to the lowest one, losing heat
- Only the lowest excited state drives photochemistry
- So count photons: measure light in moles of photons
- Red photons are used slightly more efficiently than blue



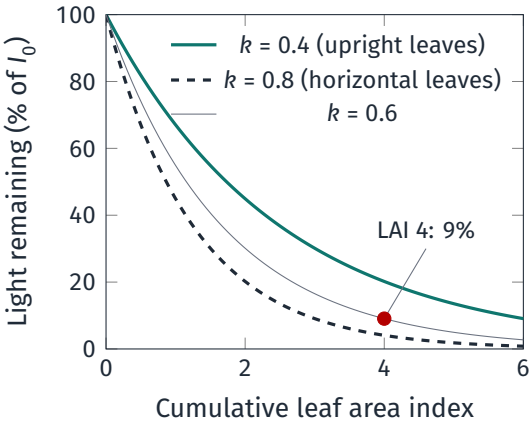
Light in the atmosphere

- Low Sun: light spread over more ground and a longer path through air
- Air molecules scatter blue strongly (Rayleigh), so the sky is blue
- Ozone absorbs UV-B; water vapour and CO_2 absorb parts of the IR
- Thick cloud can remove more than 80% of light
- **Direct** light: a beam from the Sun's disc; **diffuse** light: scattered, from all directions



Light under plant canopies

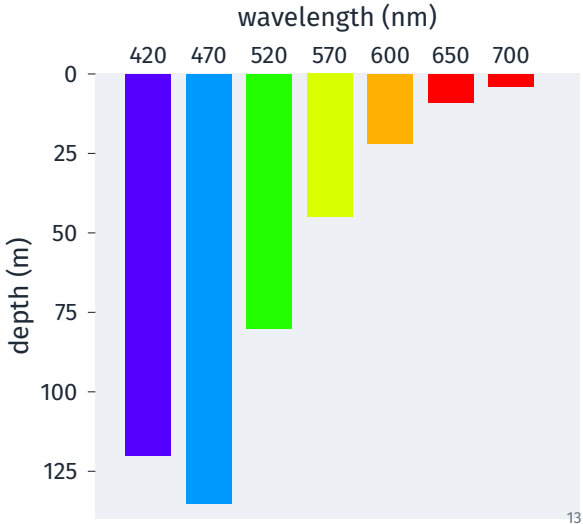
- Light declines exponentially with leaf area: $I = I_0\, e^{-k \cdot \mathrm{LAI}}$
- LAI = leaf area per unit ground area; k larger for horizontal leaves
- LAI = 4, k = 0.6: about 9% of light reaches the ground
- Leaves absorb red but pass far-red: R:FR falls from about 1.2 to below 0.3
- Low R:FR triggers shade avoidance; sunflecks supply much forest-floor light



Curves: $I/I_0 = e^{-k \cdot \mathrm{LAI}}$.

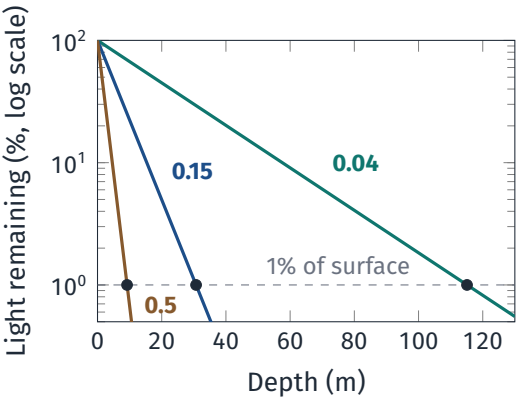
Why red light disappears first in water

- Water molecules absorb red and IR strongly (bond vibrations), blue weakly
- At 700 nm, absorption is about 40–140 times that at 420–480 nm
- Red is gone within the top few metres; blue goes deepest in clear ocean
- The sea looks blue: red absorbed, blue scattered back by water



The Beer–Lambert law in water

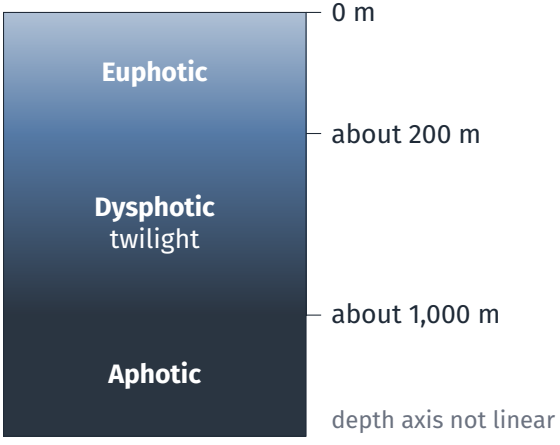
- $I_z = I_0\, e^{-K_d z}$; K_d = diffuse attenuation coefficient (m^{-1})
- From two depths: $K_d = \ln(I_1/I_2)/(z_2 - z_1)$
- Euphotic depth (1% of surface PAR): $z_{eu} = \ln(100)/K_d \approx 4.6/K_d$
- Secchi disk estimate: $K_d \approx 1.7/z_{SD}$
- An approximation: K_d differs between wavelengths and water layers



Curves: $I_z/I_0 = e^{-K_d z}$, labelled with K_d (m^{-1}). Dots: euphotic depths (9, 31, 115 m).

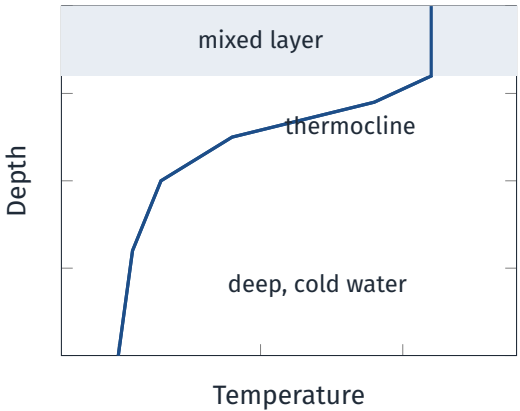
Light zones in the ocean

- **Euphotic**: surface to 1% PAR; nearly all marine photosynthesis; up to about 200 m in the clearest water, a few metres in turbid water
- **Dysphotic (twilight)**: to about 1,000 m; too little light for net photosynthesis
- **Aphotic**: below about 1,000 m; only bioluminescence



Heat from sunlight and the mixed layer

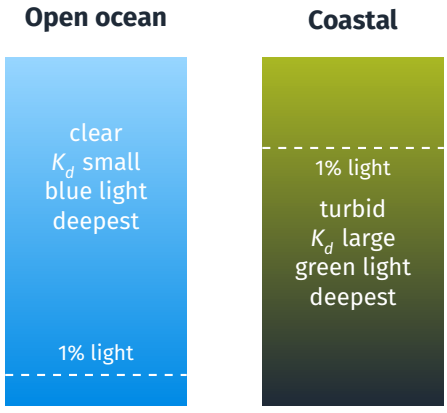
- Most solar heat is absorbed in the top metre or two (infrared and red)
- Wind and waves mix the heat down into the mixed layer
- Below it, temperature falls quickly through the thermocline
- Mixed-layer depth is set by mixing, not by light penetration
- South African west coast: upwelling brings cold deep water to the surface



Schematic; not to scale.

Coastal and open-ocean water

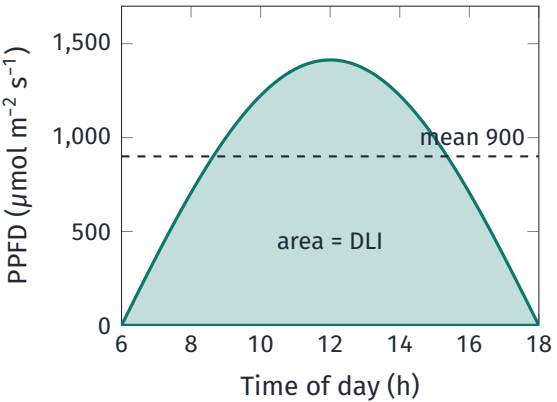
- Three groups of substances besides water attenuate light:
  - phytoplankton pigments (absorb blue and red)
  - coloured dissolved organic matter, CDOM (absorbs violet and blue)
  - suspended particles, TSS (mainly scatter)
- Coastal water: larger K_d, shallower euphotic zone
- Colour that penetrates furthest shifts from blue (ocean) to green, about 550 nm (coast)



Schematic; not to scale.

Measuring light: PPFD and the daily light integral

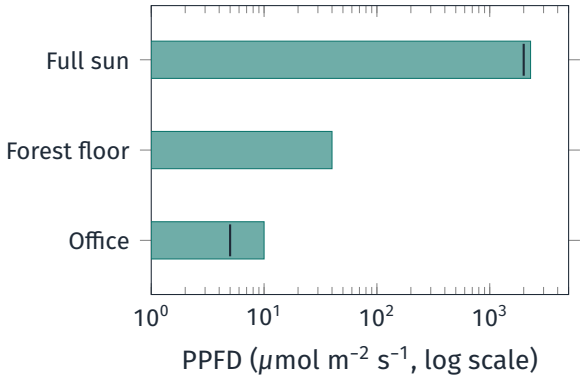
- PPFD: photons (400–700 nm) per m^2 per s, in μmol m^{-2} s^{-1}
- 1 μmol m^{-2} s^{-1} = 6.022 × 10^{17} photons m^{-2} s^{-1}
- Summed over a day: daily light integral (DLI), mol m^{-2} d^{-1}
- Mean 900 μmol m^{-2} s^{-1} for 12 h gives a DLI of 38.9 mol m^{-2} d^{-1}
- A PPFD reading says nothing about light quality



Idealised clear day: half-sine curve with a mean of 900 μmol m^{-2} s^{-1} over 12 h.

Typical light levels

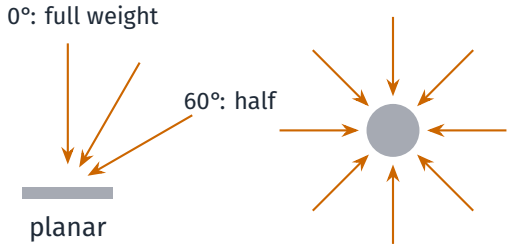
- Clear-sky midday sun at sea level: about 2,000–2,300 μmol m^{-2} s^{-1}
- Much lower in winter, early and late in the day, and under cloud
- Dense forest floor: may receive less than 2% of the light above the canopy
- A well-lit office: about 5–10 μmol m^{-2} s^{-1}

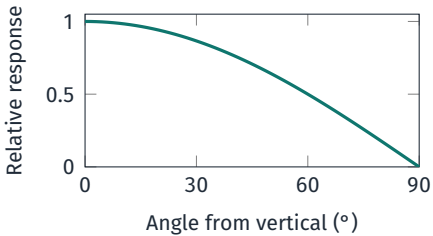


Bars show the upper value of each range; ticks the lower value. Forest floor: 2% of 2,000.

Light sensors

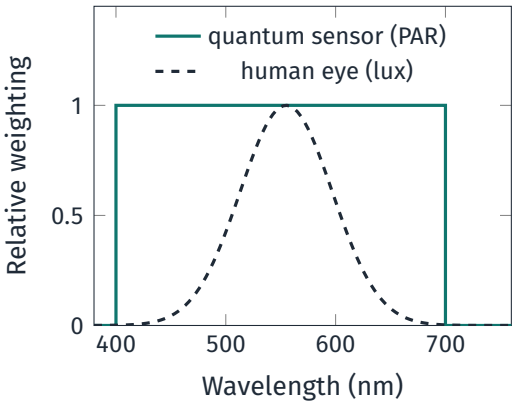
- **Cosine-corrected (planar)**: flat collector; light from the whole upper hemisphere, weighted by the cosine of its angle from vertical (like a flat leaf)
- **Spherical (scalar)**: equal response from all directions (like a phytoplankton cell)
- **Spectroradiometer**: light at each wavelength (quality)
- **Pyranometer**: total solar energy, W m^{-2}





Three kinds of light units

- **Quantum** (μmol m^{-2} s^{-1}): counts photons; the standard for photosynthesis
- **Radiometric** (W m^{-2}): energy; for energy balance and efficiency
- For sunlight: 1 W m^{-2} PAR ≈ 4.57 μmol m^{-2} s^{-1} (depends on the spectrum)
- **Photometric** (lux, foot-candles): weighted by the human eye, peak at 555 nm; no exact conversion to PPFD



Schematic. Eye curve approximated by a Gaussian centred on 555 nm.

Summary: effects of each waveband

Waveband	Wavelength	Main effects on plants
UV-B	280–315 nm	Damages DNA and photosystem II; detected by UVR8; triggers flavonoid sunscreens
UV-A	315–400 nm	Detected by cryptochromes and phototropins; little photosynthesis
PAR	400–700 nm	Drives photosynthesis; blue sensed by cryptochromes and phototropins; red by phytochrome
Far-red	700–750 nm	Detected by phytochrome (shade avoidance); aids photosynthesis via photosystem I
Infrared	>750 nm	Absorbed as heat; little photochemical effect

